**Shiv Nadar University Chennai**

**CS3807 – Deep Learning Laboratory**

**Experiments 8, 9**

**Effect of Attention Type and Attention Position in a Pretrained ResNet50**

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|---|---|---|
| **Degree & Branch** | B.Tech Artificial Intelligence & Data Science | Semester V |
| **Subject Code & Name** | CS3807 – Deep Learning Laboratory | AY: 2026–27 |

# 1. Objective

The objective of this experiment is to understand how different attention mechanisms modify a pretrained convolutional neural network and how the **position at which attention is inserted** influences image classification performance.

A single pretrained architecture,

ResNet50 pretrained on ImageNet

is used throughout the experiment.

Students will investigate:

- channel attention;
- spatial attention;
- sequential channel–spatial attention;
- lightweight self-attention;
- different positions for inserting attention within ResNet50;
- single-stage and multi-stage attention;
- attention ordering;
- additional established attention mechanisms; and
- qualitative interpretation using internal attention maps and Grad-CAM.

The experiment emphasizes that the effectiveness of attention depends not only on the attention mechanism itself, but also on the characteristics of the feature representation on which it operates.

# 2. Learning Outcomes

After completing this experiment, students will be able to:

1. explain the stage-wise architecture of a pretrained ResNet50;
2. distinguish early, intermediate, and late convolutional features;
3. explain the motivation behind attention mechanisms;
4. implement channel attention;
5. implement spatial attention;
6. implement sequential channel–spatial attention;
7. implement a lightweight self-attention mechanism;
8. insert an attention module at different stages of a pretrained network;
9. analyze the influence of attention position;
10. compare different attention mechanisms under a controlled experimental protocol;
11. analyze classification accuracy and class-wise performance;
12. visualize internal attention maps;
13. generate Grad-CAM explanations;
14. distinguish an internal attention map from a post-hoc explanation map;
15. analyze accuracy–complexity trade-offs; and
16. perform controlled ablation studies.

# 3. Dataset

**Recommended Dataset: TensorFlow Flowers (`tf_flowers`)**

The TensorFlow Flowers dataset contains natural RGB flower images belonging to five classes:

`daisy`, `dandelion`, `roses`, `sunflowers`, `tulips`.

The images contain variations in:

- scale;
- viewpoint;
- illumination;
- background clutter;
- flower size;
- color;
- texture; and
- object location.

These variations make the dataset suitable for studying the behavior of attention mechanisms.

### 3.1 Image Preprocessing

Resize every input image to

$$224 \times 224 \times 3.$$

The image should be preprocessed using the preprocessing procedure expected by the ImageNet-pretrained ResNet50.
Suitable data augmentation may include:

- random horizontal flipping;
- small rotations;
- random zoom;
- random translation; and
- mild contrast variation.

The same augmentation strategy must be maintained for all model comparisons.

### 3.2 Dataset Split

Use the same training, validation, and testing split for every experiment:

$$\boxed{70\% : 15\% : 15\%}$$

for

$$\text{Training} : \text{Validation} : \text{Testing}.$$

The exact same samples should be retained in each subset for all ablation experiments.
This is necessary to ensure that differences in performance are caused by changes in the model rather than changes in the dataset split.

## 4. Pretrained Architecture: ResNet50

ResNet50 pretrained using ImageNet is used as the **only backbone architecture** in this experiment.
For an input image

$$I \in \mathbb{R}^{224\times224\times3},$$

the major ResNet50 stages produce approximately the following feature dimensions:

| **ResNet50 Stage** | **Spatial Resolution** | **Channels** |
|---|---|---|
| Stem + MaxPool | $56 \times 56$ | 64 |
| conv2_x | $56 \times 56$ | 256 |
| conv3_x | $28 \times 28$ | 512 |
| conv4_x | $14 \times 14$ | 1024 |
| conv5_x | $7 \times 7$ | 2048 |

As the network becomes deeper:

$$\boxed{\text{Spatial Resolution} \downarrow}$$

while

$$\boxed{\text{Feature Channels} \uparrow}$$

and the representation generally progresses from low-level visual information toward higher-level semantic information.

## 5. Understanding ResNet50 Feature Hierarchy

The approximate interpretation of different stages is:

- **Early features:** edges, colors, orientations, boundaries, and simple textures.
- **Intermediate features:** local patterns, repeated structures, flower parts, petal arrangements, and shape information.
- **Late features:** high-level object structures and class-discriminative semantic information.

Therefore, the same attention mechanism may produce different behavior when inserted at different locations.

## 6. Why Attention Position Matters

Attention is not independent of the features on which it operates.
Suppose an attention module receives

$$F \in \mathbb{R}^{H \times W \times C}.$$

The values of

$$H, \qquad W, \qquad C$$

and the semantic meaning of the feature tensor change significantly with network depth.
Therefore:

- **early-stage attention** operates on high-resolution, low-level visual information;
- **middle-stage attention** operates on intermediate features containing both spatial and semantic information; and
- **late-stage attention** operates on low-resolution, high-level semantic representations.

Thus, students should investigate both:

$$\boxed{\text{Which attention mechanism should be used?}}$$

and

$$\boxed{\text{Where should the attention mechanism be inserted?}}$$

## 7. Candidate Attention Positions

Define four candidate attention positions in ResNet50.

$$P_1 : \text{ after conv2_x},$$

$$P_2 : \text{ after conv3_x},$$

$$P_3 : \text{ after conv4_x},$$

$$P_4 : \text{ after conv5_x}.$$

The corresponding feature dimensions are:

| Position | Location | Resolution | Channels |
|---|---|---|---|
| $P_1$ | After conv2_x | $56 \times 56$ | 256 |
| $P_2$ | After conv3_x | $28 \times 28$ | 512 |
| $P_3$ | After conv4_x | $14 \times 14$ | 1024 |
| $P_4$ | After conv5_x | $7 \times 7$ | 2048 |

For the principal laboratory exercise, students should concentrate on

$$\boxed{P_2, \quad P_3, \quad P_4}$$

because $P_1$ contains comparatively high-resolution features and may substantially increase the computational cost of some attention mechanisms.

# 8. Baseline Model

The baseline is the pretrained ResNet50 without any additional attention mechanism.

The baseline establishes the reference performance against which the attention-based models are compared.

# 9. Channel Attention

Consider a feature tensor

$$F \in \mathbb{R}^{H \times W \times C}.$$

Channel attention determines the relative importance of different feature channels.
Global average pooling is first used to generate a channel descriptor:

$$z_c = \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} F(i, j, c).$$

Thus,

$$z \in \mathbb{R}^C.$$

The channel descriptor is transformed using two learnable transformations:

$$M_c(F) = \sigma\left(W_2 \delta\left(W_1 z\right)\right),$$

where:

- $\delta(\cdot)$ denotes a nonlinear activation such as ReLU;
- $\sigma(\cdot)$ denotes the sigmoid activation;
- $W_1$ reduces the channel representation; and
- $W_2$ restores the channel dimension.

The refined feature tensor is:

$$F_c = F \odot M_c(F).$$

Channel attention attempts to answer:

$$\boxed{\text{Which feature channels are important?}}$$

A reduction ratio of

$$r = 16$$

may be used as the default setting throughout the laboratory experiment.
The reduction ratio need not be treated as a separate mandatory ablation.

## 10. Spatial Attention

Spatial attention determines the spatial regions that should receive greater importance.
For the input feature tensor

$$F \in \mathbb{R}^{H\times W\times C},$$

compute average pooling across channels:

$$F_{\text{avg}} = \text{Mean}_c(F),$$

and maximum pooling across channels:

$$F_{\max} = \text{Max}_c(F).$$

Both tensors have dimensions

$$H \times W \times 1.$$

Concatenate them:

$$F_s = [F_{\text{avg}} \| F_{\max}].$$

The spatial attention map is then computed as

$$M_s(F) = \sigma\left(\text{Conv}_{7\times 7}(F_s)\right).$$

Therefore,

$$M_s(F) \in \mathbb{R}^{H\times W\times 1}.$$

The spatially refined feature is

$$F'_s = F \odot M_s(F).$$

Spatial attention answers:

$$\boxed{\text{Which spatial locations are important?}}$$

## 11. Sequential Channel–Spatial Attention

Channel and spatial attention can be sequentially applied.
First,

$$F_c = M_c(F) \odot F.$$

Then,

$$F_{cs} = M_s(F_c) \odot F_c.$$

The overall structure becomes:

$$\boxed{F \to \text{Channel Attention} \to \text{Spatial Attention} \to F_{cs}}$$

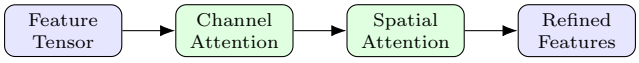
This arrangement first determines

"what is important"

and then determines

"where it is important."

A simplified diagram is shown below.



# 12. Self-Attention

Self-attention allows spatial locations to interact directly with other spatial locations.
For

$$F \in \mathbb{R}^{H \times W \times C},$$

flatten the spatial dimensions:

$$X \in \mathbb{R}^{N \times C},$$

where

$$N = HW.$$

Generate the query, key, and value matrices:

$$Q = XW_Q,$$

$$K = XW_K,$$

and

$$V = XW_V.$$

The attention matrix is

$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right),$$

where $d$ is the dimension of the query/key representation.
The attended output is

$$Y = AV.$$

After reshaping the output back to the original spatial structure, a residual formulation may be used:

$$F_{\text{SA}} = F + \gamma\,\text{reshape}(Y),$$

where $\gamma$ is a learnable scaling parameter.
Self-attention attempts to answer:

| Which spatial features are related to other spatial features? |
|---|

Since the attention matrix has dimension approximately

$$N \times N,$$

its computational requirements increase rapidly as

$$N = HW$$

increases.
Therefore, self-attention should preferably be introduced at

$$P_3$$

or

$$P_4.$$

# 13. Inserting Attention into ResNet50

For any selected stage $P_k$, the network structure can be written as

$$\text{Previous ResNet Stage} \rightarrow \text{Attention} \rightarrow \text{Next ResNet Stage}.$$

For example, attention inserted after conv4_x gives:

$$\boxed{\text{conv4_x} \rightarrow \text{Attention} \rightarrow \text{conv5_x}}$$

For channel–spatial attention:

$$\boxed{\text{conv4_x} \rightarrow \text{Channel Attention} \rightarrow \text{Spatial Attention} \rightarrow \text{conv5_x}}$$

The student must record the feature dimensions immediately before and after the attention block.
Attention should not unintentionally modify the spatial or channel dimensions unless the selected mechanism explicitly requires such a transformation.

# 14. Main Experiment 1: Attention-Type Ablation

The first experiment compares the baseline ResNet50 with attention-based models while keeping the attention position fixed at $P_3$, after `conv4_x`. The same five-class TensorFlow Flowers task is used for each model.

| Model | Accuracy | Precision | Recall | Macro F1 | Parameters | Inference (ms/image) |
|---|---|---|---|---|---|---|
| Baseline ResNet50 | 0.9347 | 0.9350 | 0.9332 | 0.9339 | 23,597,957 | 7.6947 |
| Channel Attention | 0.9401 | 0.9409 | 0.9376 | 0.9389 | 23,730,117 | 7.8742 |
| Spatial Attention | 0.9292 | 0.9275 | 0.9293 | 0.9283 | 23,598,056 | 7.9831 |
| Channel+Spatial | 0.9201 | 0.9196 | 0.9206 | 0.9194 | 23,730,216 | 7.9179 |
| Self-Attention | 0.9347 | 0.9360 | 0.9331 | 0.9343 | 24,909,958 | 8.2831 |
| Combined Attention | 0.9274 | 0.9258 | 0.9283 | 0.9267 | 25,042,217 | 8.3501 |

**Figure 1: Attention Mechanism versus Accuracy**

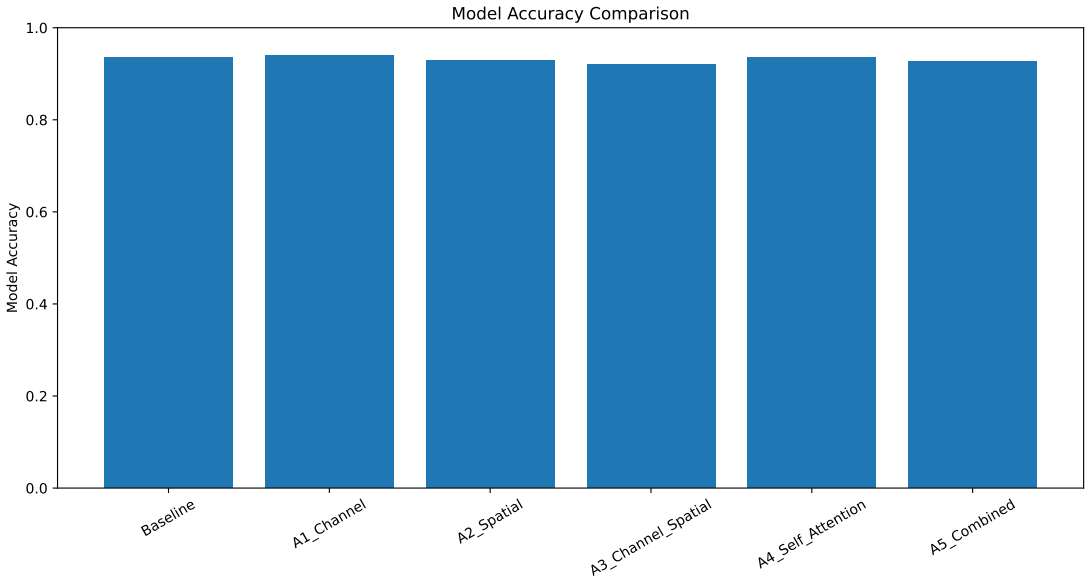
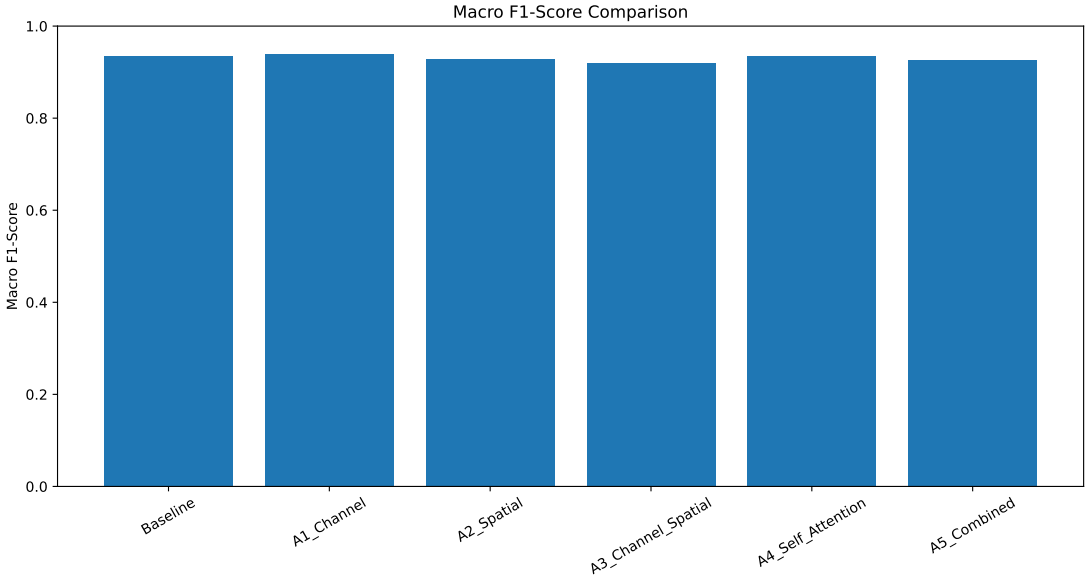



**Figure 2: Attention Mechanism versus Macro F1-score**



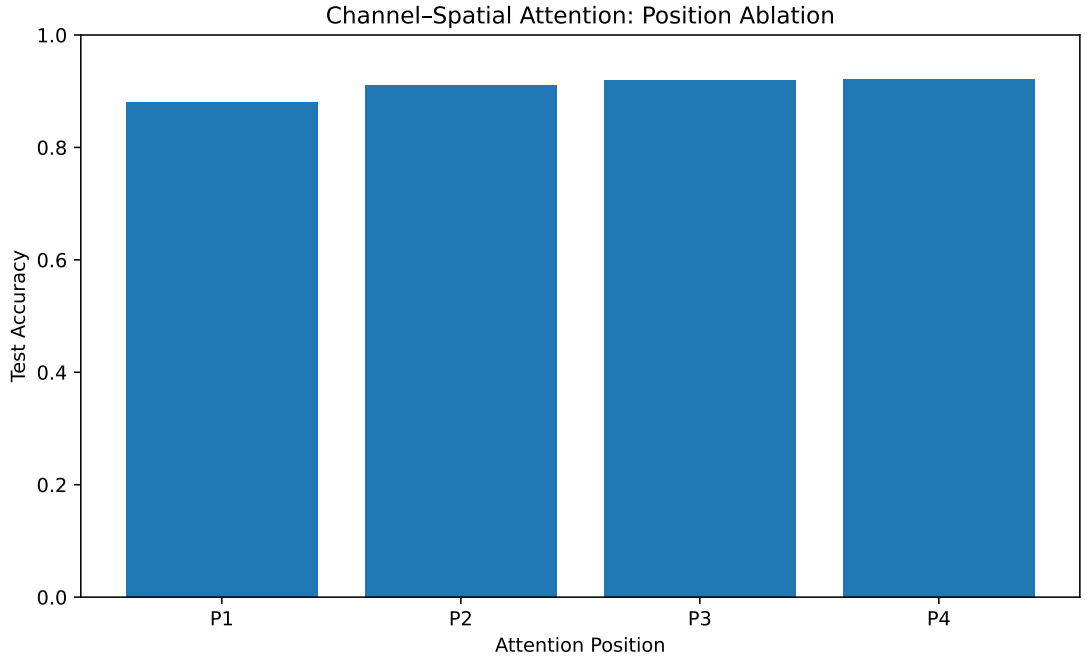

**Observation:**
In these runs, channel attention obtained the highest test accuracy (0.9401) and macro F1-score (0.9389). Its performance was slightly higher than the baseline accuracy of 0.9347. Spatial attention, channel–spatial attention, and combined attention did not outperform the baseline in this comparison. This result shows that adding more attention operations does not guarantee an improvement for every model or dataset.

# 15. Main Experiment 2: Attention-Position Ablation

In this experiment, the same sequential channel–spatial attention module was inserted at different ResNet50 stages. Only the insertion position was intended to change.

| Position | Feature Size | Accuracy | Precision | Recall | Macro F1 | Inference (ms/image) |
|---|---|---|---|---|---|---|
| $P_1$ | $56 \times 56 \times 256$ | 0.8802 | 0.8855 | 0.8780 | 0.8807 | 8.0656 |
| $P_2$ | $28 \times 28 \times 512$ | 0.9111 | 0.9090 | 0.9151 | 0.9106 | 7.8956 |
| $P_3$ | $14 \times 14 \times 1024$ | 0.9201 | 0.9196 | 0.9206 | 0.9194 | 8.0353 |
| $P_4$ | $7 \times 7 \times 2048$ | 0.9220 | 0.9196 | 0.9222 | 0.9201 | 8.1700 |

**Figure 3: Attention Position versus Accuracy**



**Observation:**

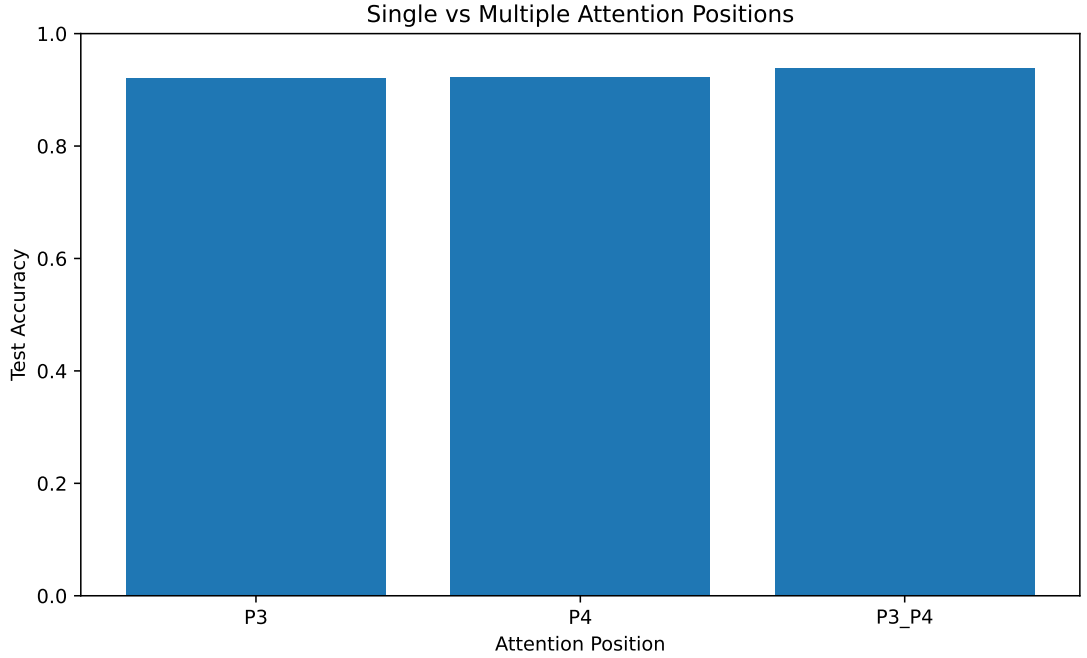
Among the four tested positions, $P_4$ produced the highest recorded accuracy (0.9220), closely followed by $P_3$ (0.9201). The lowest accuracy was observed at $P_1$ (0.8802). The results show that the position of the attention block affected the model's performance in this experiment. They do not imply that the deepest position must always be best for every task.

# 16. Main Experiment 3: Multiple Attention Positions

This experiment examines whether using attention at more than one stage provides a sufficient improvement to justify the extra model complexity. The notebook compared attention at $P_3$, attention at $P_4$, and attention at both $P_3 + P_4$.

| Configuration | Test Accuracy |
|---|---|
| Attention at $P_3$ | 0.9201 |
| Attention at $P_4$ | 0.9220 |
| Attention at $P_3 + P_4$ | 0.9383 |

**Figure 4: Multiple Attention Positions**




**Observation:**

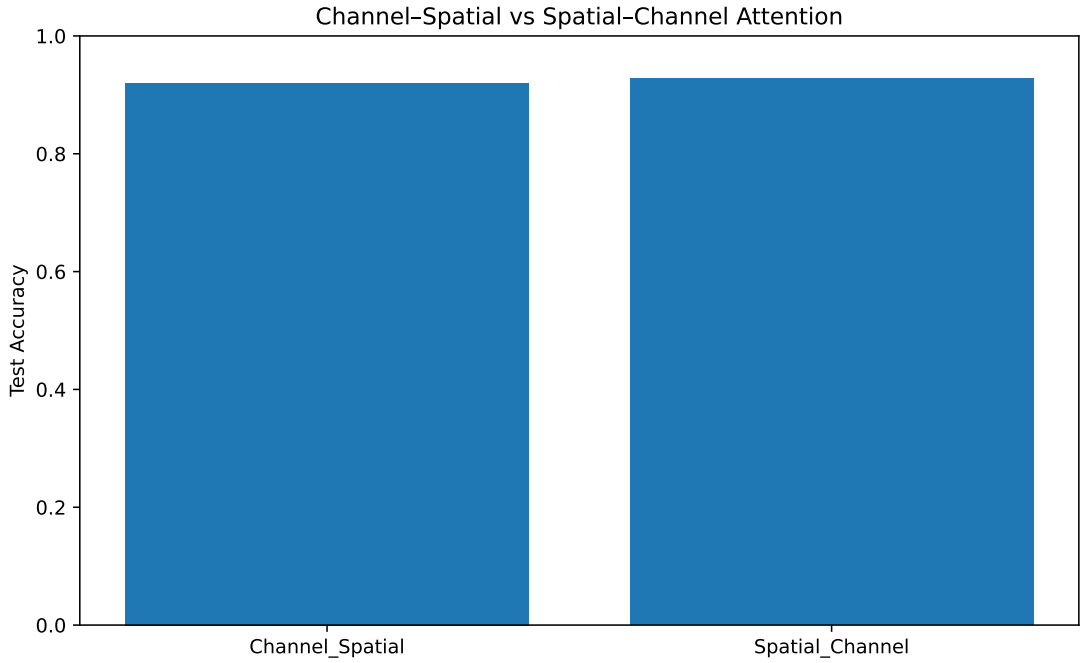
The $P_3 + P_4$ configuration achieved an accuracy of 0.9383, higher than the single-position values reported above. For this run, combining the two stages was beneficial in terms of accuracy. However, the extra computation, parameters, and inference cost should also be considered before deciding that a multi-stage model is preferable.

# 17. Main Experiment 4: Attention Order

The order of channel and spatial attention was compared while maintaining the same attention position and the same general training protocol.

| Attention Order | Accuracy | Macro F1 |
|---|---|---|
| Channel → Spatial | 0.9201 | 0.9194 |
| Spatial → Channel | 0.9274 | 0.9267 |

**Figure 5: Attention Order Comparison**




**Observation:**

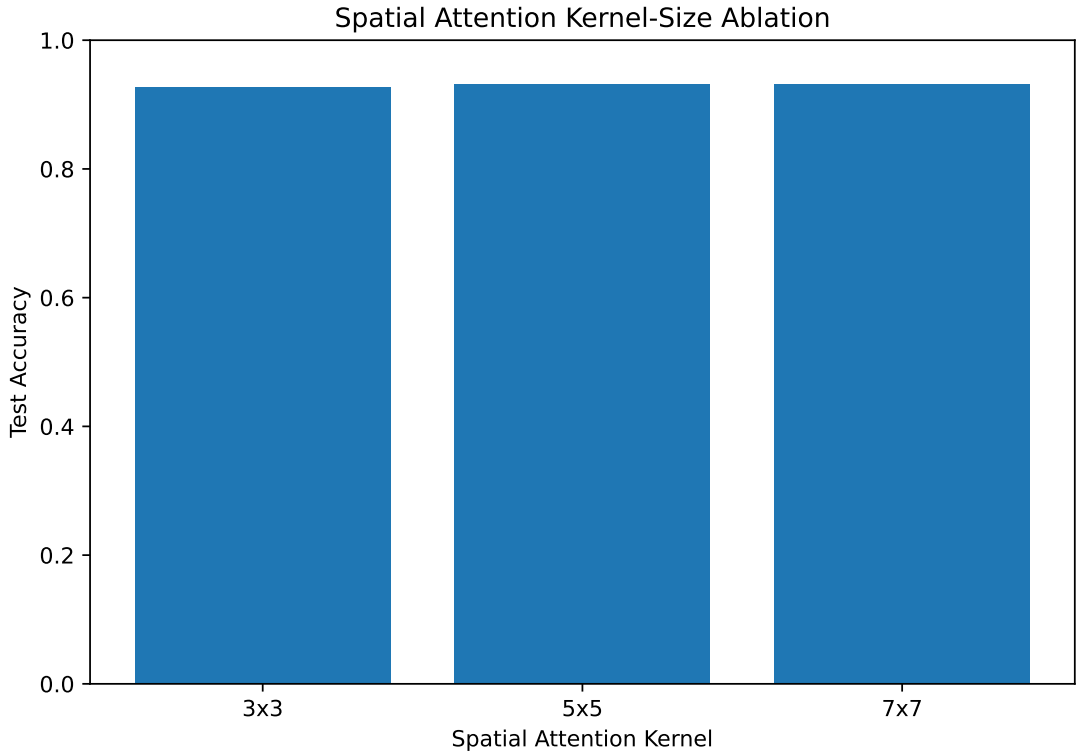
Spatial–channel attention achieved higher accuracy than channel–spatial attention in the recorded comparison. The measured result suggests that the order of the two operations can affect the learned representation. This conclusion is limited to the tested model and training run.

# 18. Main Experiment 5: Spatial Attention Kernel Size

The spatial-attention kernel size controls the receptive field used to calculate the spatial attention map. The experiment compared $k \in \{3, 5, 7\}$.

| Spatial Kernel | Accuracy | Macro F1 |
|---|---|---|
| $3 \times 3$ | 0.9274 | 0.9263 |
| $5 \times 5$ | 0.9310 | 0.9304 |
| $7 \times 7$ | 0.9310 | 0.9304 |

**Figure 6: Spatial Kernel Size Comparison**




**Observation:**
The $5 \times 5$ and $7 \times 7$ kernels obtained the same recorded accuracy and macro F1-score, both slightly higher than the $3 \times 3$ kernel. In this experiment, increasing the kernel from $5 \times 5$ to $7 \times 7$ did not produce a further measured improvement.

# 19. Additional Task: Comparison of Different Attention Mechanisms

The additional comparison uses the same ResNet50 backbone and places the established attention mechanisms at $P_3$ where applicable. The following results were recorded by the notebook.

| Attention | Accuracy | Precision | Recall | Macro F1 | Parameters |
|---|---|---|---|---|---|
| No Attention | 0.9347 | 0.9350 | 0.9332 | 0.9339 | 23,597,957 |
| Channel Attention | 0.9401 | 0.9409 | 0.9376 | 0.9389 | 23,730,117 |
| Spatial Attention | 0.9292 | 0.9275 | 0.9293 | 0.9283 | 23,598,056 |
| Channel+Spatial | 0.9201 | 0.9196 | 0.9206 | 0.9194 | 23,730,216 |
| Self-Attention | 0.9347 | 0.9360 | 0.9331 | 0.9343 | 24,909,958 |
| SE | 0.9310 | 0.9300 | 0.9312 | 0.9302 | 23,730,117 |
| ECA | 0.9201 | 0.9191 | 0.9226 | 0.9201 | 23,597,960 |
| CBAM | 0.9238 | 0.9246 | 0.9220 | 0.9228 | 23,730,216 |

**Figure 7: Additional Attention Mechanisms**

**Observation:**
Channel attention recorded the highest accuracy among the listed models. SE and CBAM did not exceed the baseline accuracy in this run, while ECA had a parameter count close to the baseline. These results show that model complexity alone does not determine classification performance.

### 19.1 Suggested Comparison: Main Results Table

The following table combines the main attention-type and attention-position results. Inference time is given in milliseconds per image.

| Model | Position | Accuracy | Precision | Recall | Macro F1 | Parameters |
|---|---|---|---|---|---|---|
| Baseline | – | 0.9347 | 0.9350 | 0.9332 | 0.9339 | 23,597,957 |
| Channel Attention | $P_3$ | 0.9401 | 0.9409 | 0.9376 | 0.9389 | 23,730,117 |
| Spatial Attention | $P_3$ | 0.9292 | 0.9275 | 0.9293 | 0.9283 | 23,598,056 |
| Channel+Spatial | $P_2$ | 0.9111 | 0.9090 | 0.9151 | 0.9106 | 23,631,368 |
| Channel+Spatial | $P_3$ | 0.9201 | 0.9196 | 0.9206 | 0.9194 | 23,730,216 |
| Channel+Spatial | $P_4$ | 0.9220 | 0.9196 | 0.9222 | 0.9201 | 24,124,520 |
| Self-Attention | $P_3$ | 0.9347 | 0.9360 | 0.9331 | 0.9343 | 24,909,958 |
| SE | $P_3$ | 0.9310 | 0.9300 | 0.9312 | 0.9302 | 23,730,117 |
| ECA | $P_3$ | 0.9201 | 0.9191 | 0.9226 | 0.9201 | 23,597,960 |

**Inference time recorded by the notebook:**

| Model | Inference Time (ms/image) |
|---|---|
| Baseline | 7.5673 |
| Channel Attention | 7.6675 |
| Spatial Attention | 7.6061 |
| Channel+Spatial at $P_2$ | 7.8450 |
| Channel+Spatial at $P_3$ | 7.6711 |
| Channel+Spatial at $P_4$ | 7.7778 |
| Self-Attention | 7.8668 |
| SE | 7.5926 |
| ECA | 7.8403 |

| Model | Position | Accuracy | Precision | Recall | F1 | Parameters | Inference Time (ms/image) |
|---|---|---|---|---|---|---|---|
| Baseline | — | 0.9347 | 0.9350 | 0.9332 | 0.9339 | 23597957 | 7.567 |
| Channel Attention | P3 | 0.9401 | 0.9409 | 0.9376 | 0.9389 | 23730117 | 7.668 |
| Spatial Attention | P3 | 0.9292 | 0.9275 | 0.9293 | 0.9283 | 23598056 | 7.606 |
| Channel+Spatial | P2 | 0.9111 | 0.9090 | 0.9151 | 0.9106 | 23631368 | 7.845 |
| Channel+Spatial | P3 | 0.9201 | 0.9196 | 0.9206 | 0.9194 | 23730216 | 7.671 |
| Channel+Spatial | P4 | 0.9220 | 0.9196 | 0.9222 | 0.9201 | 24124520 | 7.778 |
| Self-Attention | P3 | 0.9347 | 0.9360 | 0.9331 | 0.9343 | 24909958 | 7.867 |
| Additional Attention-1 (SE) | P3 | 0.9310 | 0.9300 | 0.9312 | 0.9302 | 23730117 | 7.593 |
| Additional Attention-2 (ECA) | P3 | 0.9201 | 0.9191 | 0.9226 | 0.9201 | 23597960 | 7.840 |

### 19.2 Questions for Attention-Mechanism Comparison

1. **How does classification accuracy vary across attention mechanisms?** In the recorded attention-type comparison, channel attention achieved the highest accuracy (0.9401), while channel–spatial attention achieved 0.9201.
2. **How does macro F1-score vary?** Channel attention had the highest macro F1-score (0.9389) among the six primary attention-type entries.
3. **How many parameters are introduced?** The recorded total parameter counts range from $23,597,957$ for the baseline to $25,042,217$ for the combined-attention model.
4. **How does attention influence inference time?** The measured inference times vary across configurations. The combined attention model recorded 8.3501 ms/image in the attention-type comparison, compared with 7.6947 ms/image for the baseline.
5. **Does additional complexity always improve performance?** No. The combined-attention model had more parameters but did not exceed the baseline accuracy in this run.
6. **Can lightweight attention be competitive?** Yes. Channel attention used fewer parameters than the combined-attention model and achieved the highest accuracy in the primary comparison.

## 20. Transfer-Learning Strategy

The experiment used an ImageNet-pretrained ResNet50 backbone. The original ImageNet classification layer was replaced by a task-specific classification head for the five flower classes. The transfer-learning process followed two

stages: first, the pretrained backbone was frozen while the new layers were trained; then the final ResNet50 stage was unfrozen for fine-tuning with a smaller learning rate.

The same backbone, dataset split, image preprocessing, classifier, and evaluation procedure were maintained for the controlled comparisons. The notebook used Adam and sparse categorical cross-entropy for the five-class labels.

# 21. Loss Function

The model predicts one of five flower classes. The classification loss is cross-entropy. For $N$ samples and $C = 5$ classes, categorical cross-entropy can be written as

$$\mathcal{L}_{CE} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{5} y_{i,c}\log(\hat{y}_{i,c}),$$

where $y_{i,c}$ is the target indicator and $\hat{y}_{i,c}$ is the predicted probability for class $c$. In the notebook, the labels were handled using sparse categorical cross-entropy, which is the equivalent form for integer-encoded class labels.

# 22. Evaluation Metrics

The models were evaluated using accuracy, macro precision, macro recall, macro F1-score, parameter count, and average inference time per image.

For a class $c$,

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c}, \qquad \text{Recall}_c = \frac{TP_c}{TP_c + FN_c}.$$

The class-wise F1-score is

$$F1_c = 2\frac{\text{Precision}_c\,\text{Recall}_c}{\text{Precision}_c + \text{Recall}_c},$$

and the macro F1-score is

$$\text{MacroF1} = \frac{1}{5}\sum_{c=1}^{5} F1_c.$$

Macro-averaged metrics give equal weight to each flower class, whereas accuracy summarizes the fraction of all test samples classified correctly.

# 23. Required Main Result Table

The principal model comparison is presented in Section 19.1. The notebook-generated table is also included below as a figure.

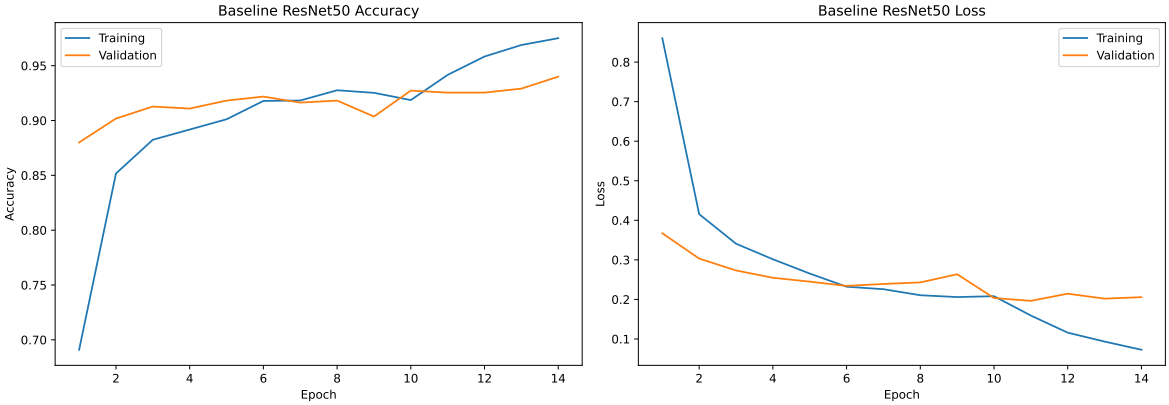
| Model | Position | Accuracy | Precision | Recall | F1 | Parameters | Inference Time (ms/image) |
|---|---|---|---|---|---|---|---|
| Baseline | — | 0.9347 | 0.9350 | 0.9332 | 0.9339 | 23597957 | 7.567 |
| Channel Attention | P3 | 0.9401 | 0.9409 | 0.9376 | 0.9389 | 23730117 | 7.668 |
| Spatial Attention | P3 | 0.9292 | 0.9275 | 0.9293 | 0.9283 | 23598056 | 7.606 |
| Channel+Spatial | P2 | 0.9111 | 0.9090 | 0.9151 | 0.9106 | 23631368 | 7.845 |
| Channel+Spatial | P3 | 0.9201 | 0.9196 | 0.9206 | 0.9194 | 23730216 | 7.671 |
| Channel+Spatial | P4 | 0.9220 | 0.9196 | 0.9222 | 0.9201 | 24124520 | 7.778 |
| Self-Attention | P3 | 0.9347 | 0.9360 | 0.9331 | 0.9343 | 24909958 | 7.867 |
| Additional Attention-1 (SE) | P3 | 0.9310 | 0.9300 | 0.9312 | 0.9302 | 23730117 | 7.593 |
| Additional Attention-2 (ECA) | P3 | 0.9201 | 0.9191 | 0.9226 | 0.9201 | 23597960 | 7.840 |

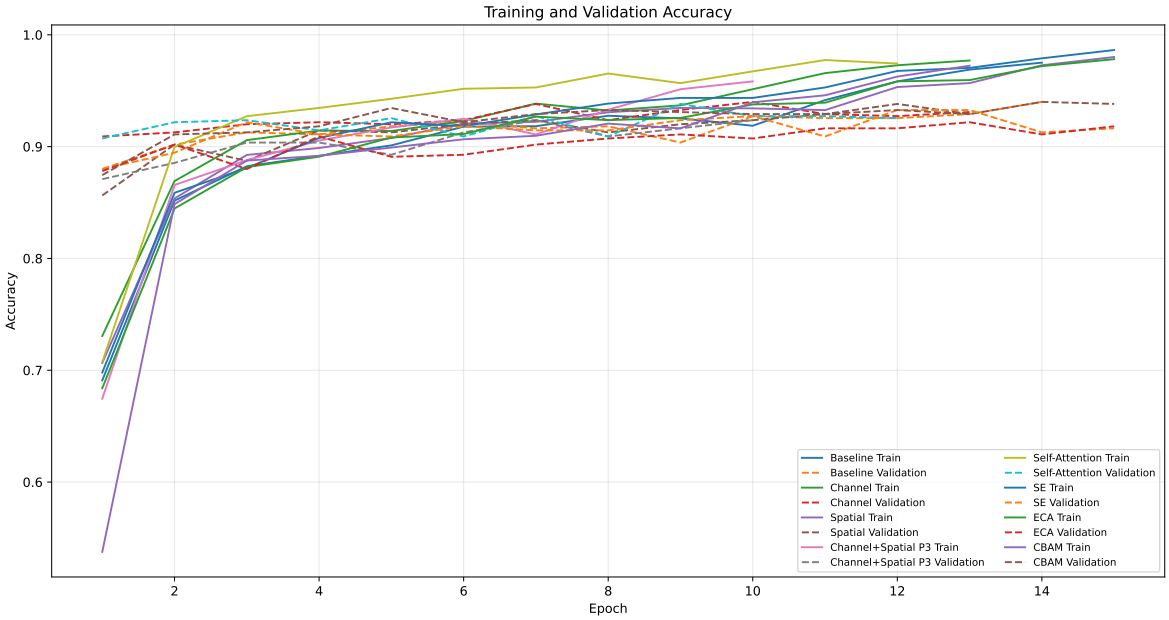
The table indicates that channel attention achieved the highest accuracy in the primary attention-type comparison. The position study and the additional mechanism comparison are reported separately because they investigate different experimental factors.

# 24. Graphical Performance Analysis

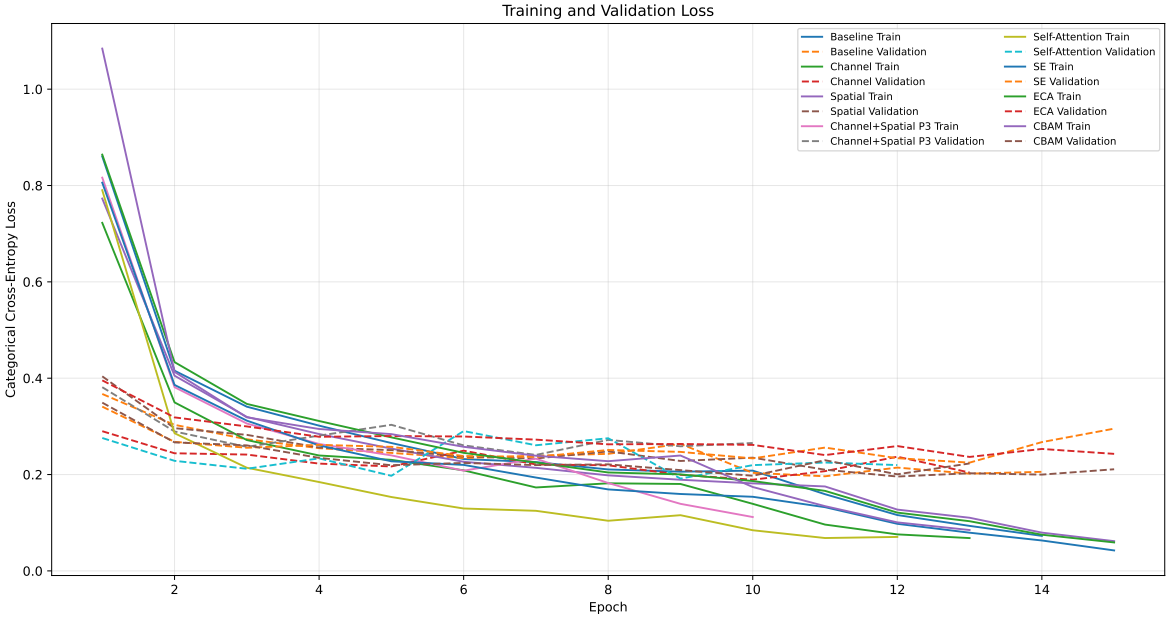
The following plots summarize performance differences between the models. The training and validation plots are included to complement the test metrics and to show the observed learning behaviour.

### 24.1 Training and Validation Accuracy

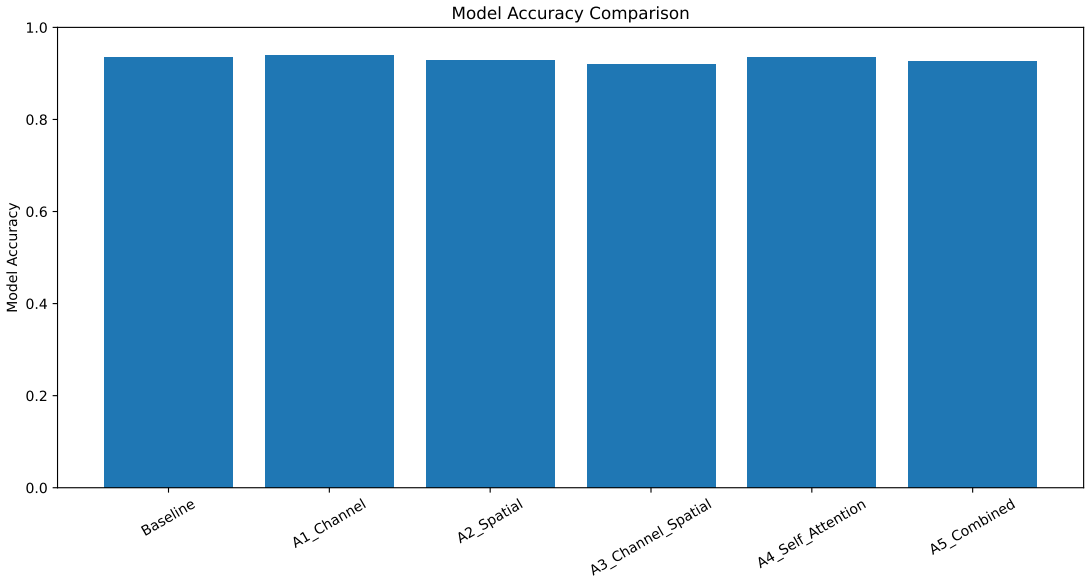


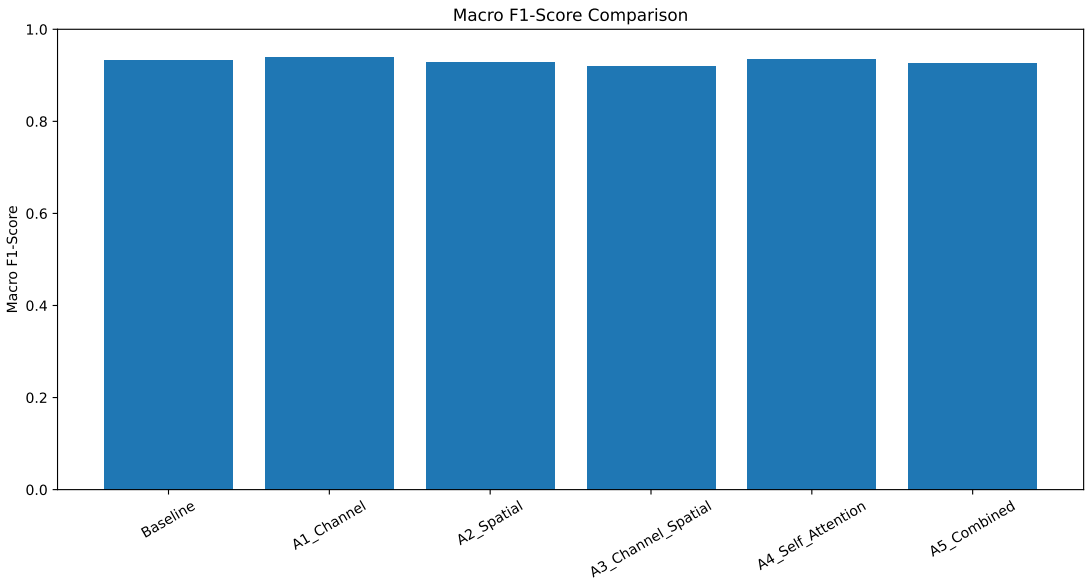

### 24.2 Training and Validation Loss



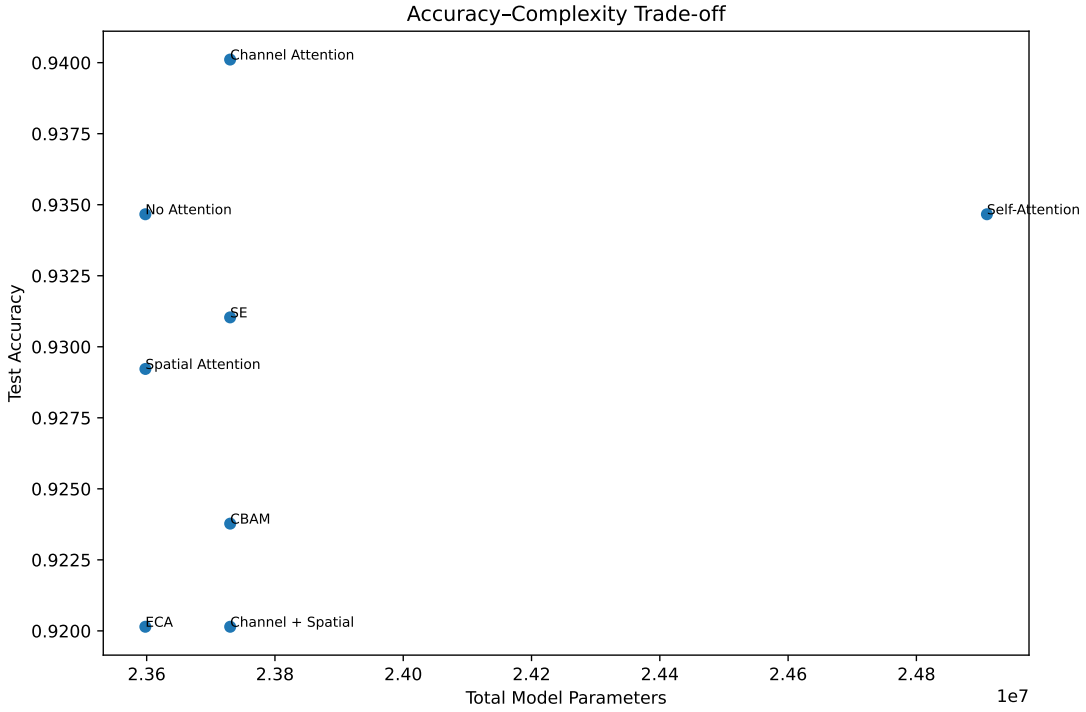
### 24.3 Attention Mechanism versus Accuracy



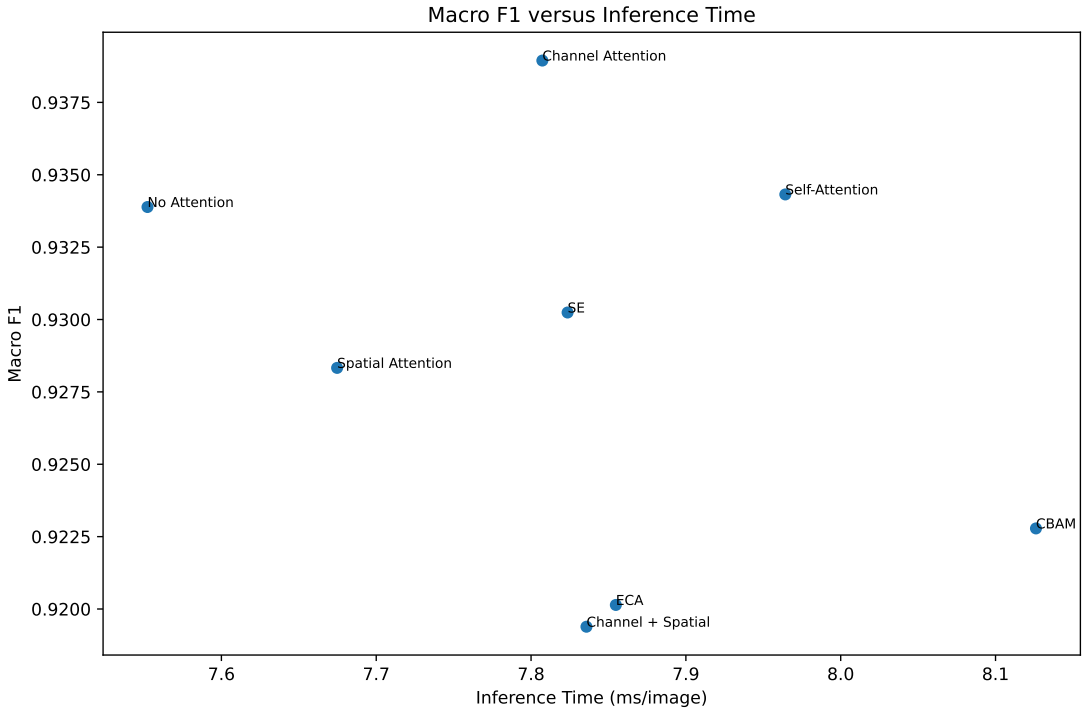

### 24.4 Attention Mechanism versus Macro F1




### 24.5 Accuracy versus Model Parameters




### 24.6 Macro F1 versus Inference Time

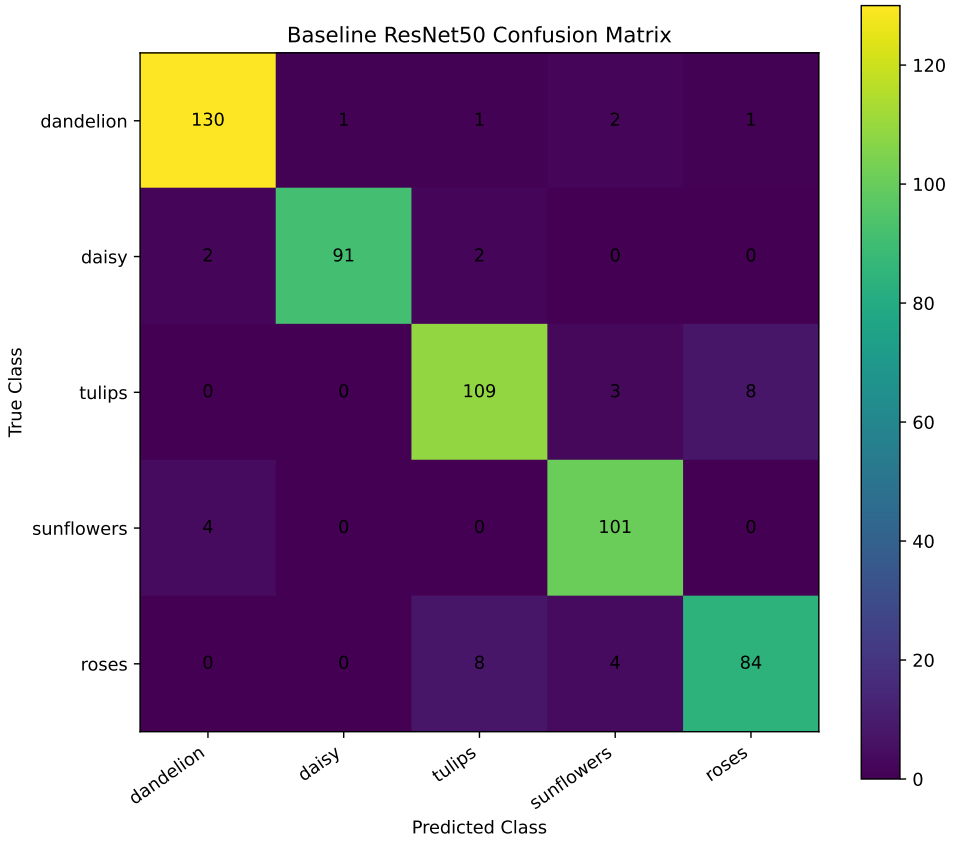



The accuracy and macro F1 plots show that the model with the most attention operations was not necessarily the best-performing model. The parameter and inference-time plots provide a second view of the trade-off between performance and computational cost.

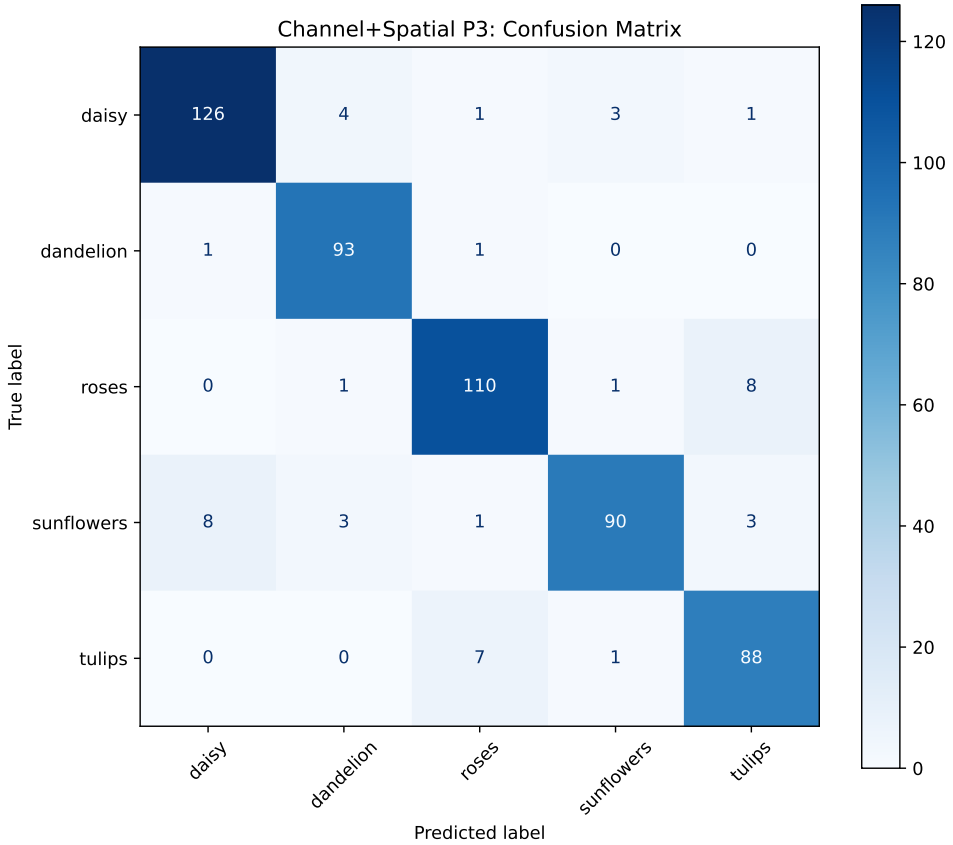
## 25. Confusion Matrix Analysis

The confusion matrices show the predicted class against the true class for the five flower categories. The baseline matrix is compared with the channel–spatial attention model at $P_3$.

**Baseline ResNet50**




**Channel–Spatial Attention at $P_3$**

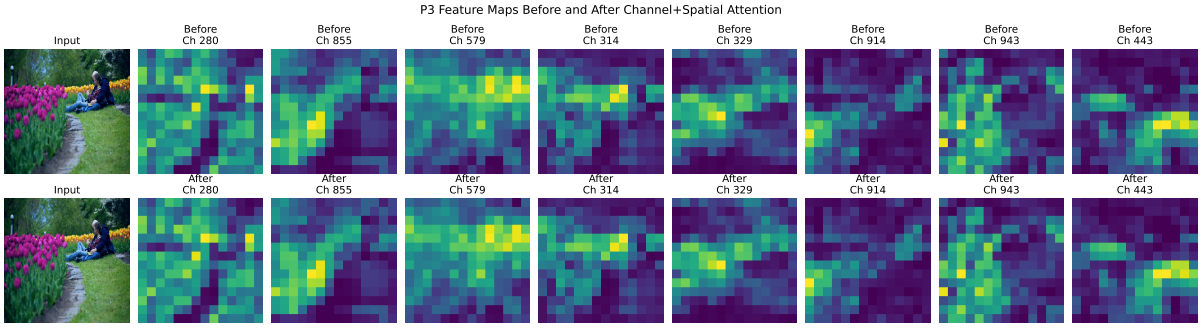



The diagonal entries represent correct classifications, while off-diagonal entries represent misclassifications. The confusion matrices should be read class by class to identify which flower categories are most often confused. The overall accuracy alone does not show the direction of these errors.

# 26. Qualitative Analysis of Attention

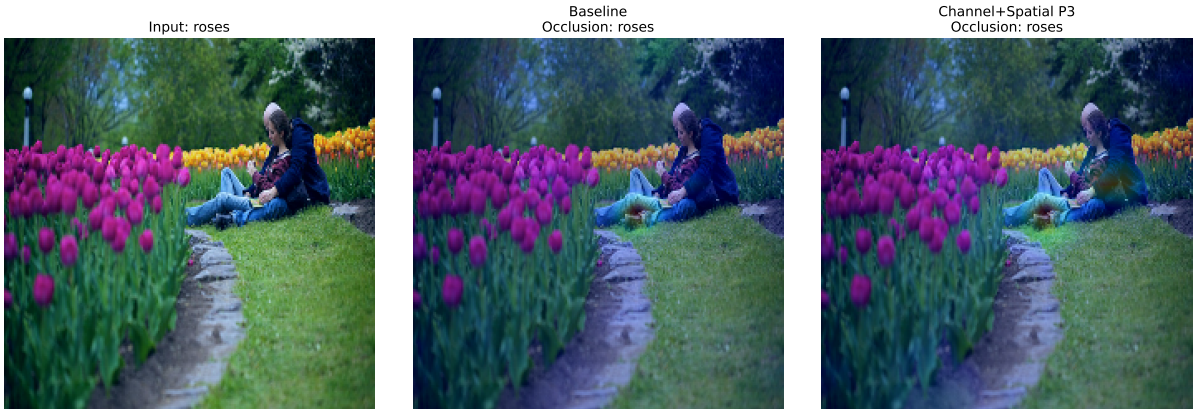
The qualitative analysis includes feature maps before and after attention, channel weights, spatial-attention overlays, self-attention responses, and Grad-CAM explanations. These visualizations provide different kinds of information and should not be treated as interchangeable.

**Feature Maps Before and After Attention**




The displayed feature maps allow a visual comparison of the representation before and after the attention operation. For the recorded $P_3$ block, the feature dimensions remain $14 \times 14 \times 1024$ before and after attention. The attention module therefore reweights the representation without changing its shape.

**Occlusion Sensitivity Comparison**



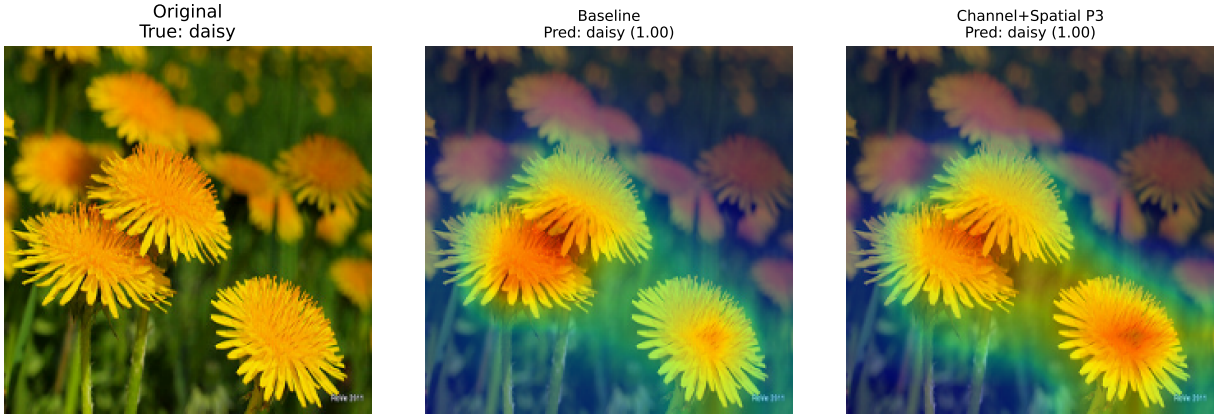

Occlusion sensitivity provides a complementary check by showing how the prediction responds when parts of an image are hidden. Interpretation should be based on the regions visible in the figure and should not assume that every high-response region corresponds to the flower itself.

# 27. Grad-CAM Visualization

Grad-CAM produces a class-specific heatmap using the gradients of the predicted class score with respect to a convolutional feature map. The baseline and attention-model visualizations are compared using the figures generated by the notebook.

**Baseline Grad-CAM**

### Grad-CAM Across Models




### Correctly Classified Examples

Grad-CAM: images correctly classified by both models

Original
True: daisy

Baseline
Pred: daisy (1.00)

Channel+Spatial P3
Pred: daisy (1.00)

Original
True: dandelion

Baseline
Pred: dandelion (1.00)

Channel+Spatial P3
Pred: dandelion (1.00)

Original
True: roses

Baseline
Pred: roses (1.00)

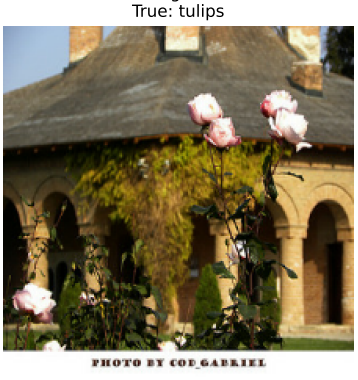
Channel+Spatial P3
Pred: roses (1.00)

Original
True: sunflowers

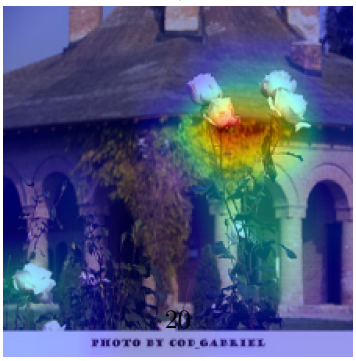
Baseline
Pred: sunflowers (1.00)

Channel+Spatial P3
Pred: sunflowers (0.96)

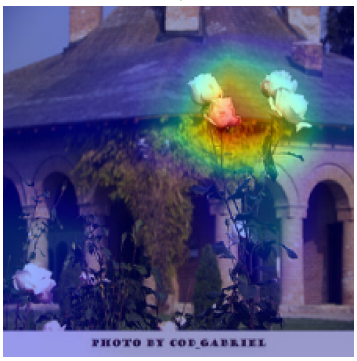
Original
True: tulips




Baseline
Pred: tulips (0.92)




Channel+Spatial P3
Pred: tulips (0.95)



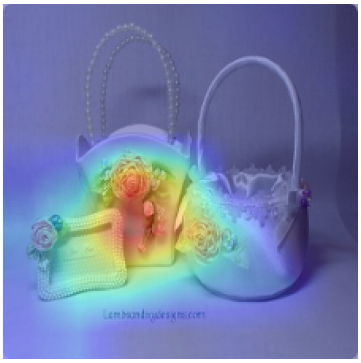
**Examples with Disagreeing Predictions**

Grad-CAM: baseline vs attention disagreements / failures

Original
True: tulips

Baseline
Pred: roses (0.57)

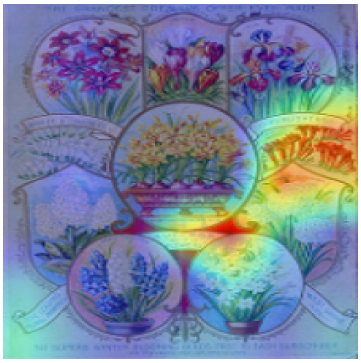
Channel+Spatial P3
Pred: tulips (0.73)



Original
True: tulips

Baseline
Pred: sunflowers (0.92)

Channel+Spatial P3
Pred: tulips (0.55)



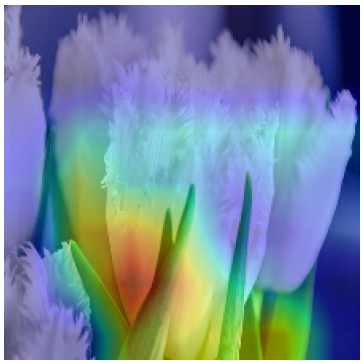
Original
True: tulips

Baseline
Pred: roses (0.68)

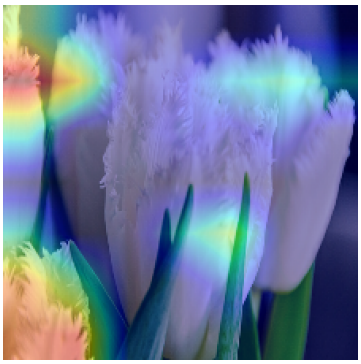
Channel+Spatial P3
Pred: tulips (0.50)

Original
True: roses

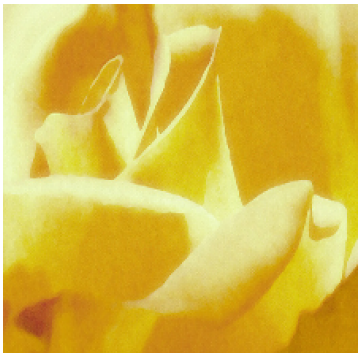
Baseline
Pred: roses (0.95)



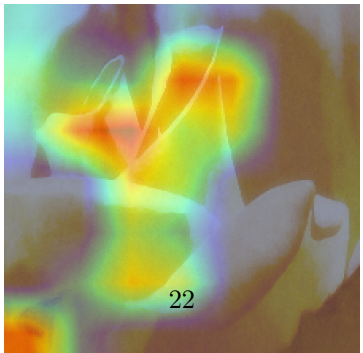
Channel+Spatial P3
Pred: dandelion (0.49)



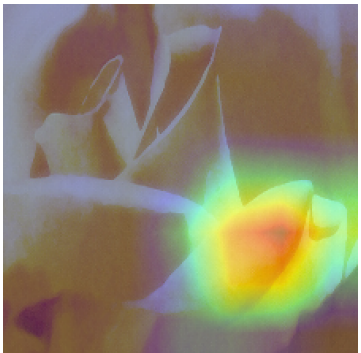
Original
True: tulips



Baseline
Pred: tulips (0.52)



Channel+Spatial P3
Pred: roses (0.93)

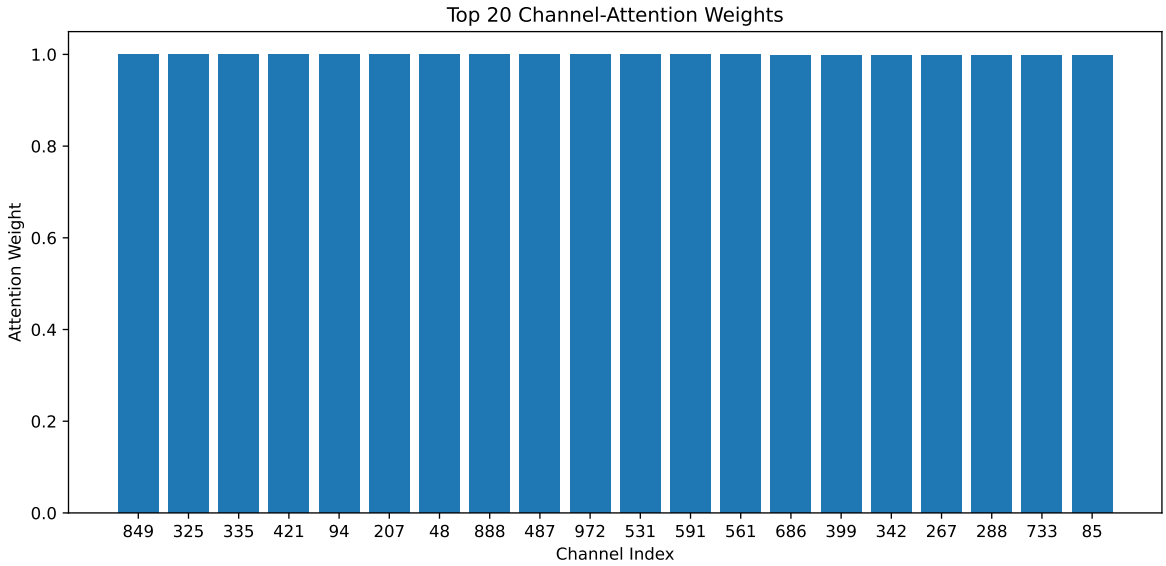


The heatmaps should be interpreted by checking which parts of each image are highlighted, including petals, the flower centre, leaves, and background. Grad-CAM is a post-hoc explanation because it is calculated after a prediction using gradients; it is distinct from an attention map computed internally during the forward pass.

# 28. Internal Attention Maps

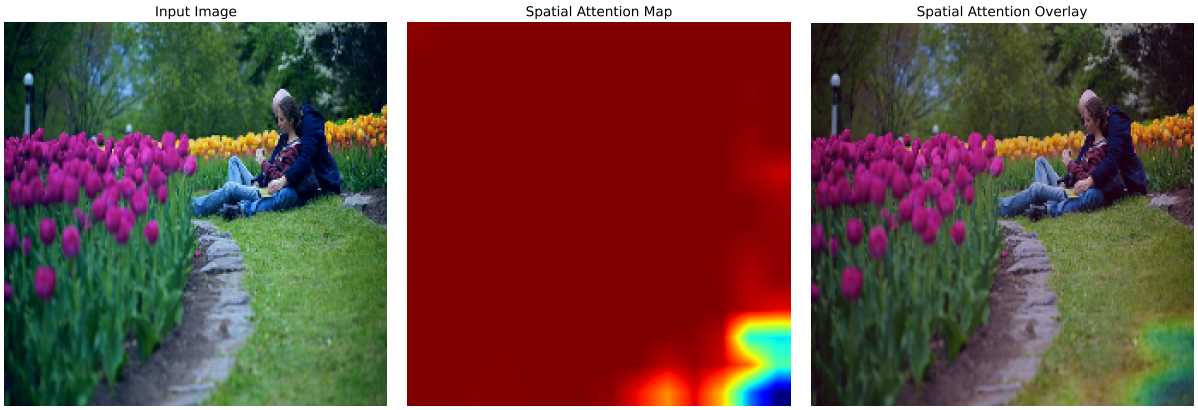
Internal attention maps are produced by the attention modules during the forward pass. They show the weights generated by the module, whereas Grad-CAM shows class-specific evidence based on gradients.

## 28.1 Channel-Attention Visualization




The top-20 channel weights are shown rather than every channel. The plot can be examined to see whether the learned weights are concentrated on a small number of channels or distributed across many channels.

## 28.2 Spatial-Attention Visualization




The spatial-attention figure displays the attention response and its relationship to the input image. Observations about foreground focus or background suppression should be based on the visible overlay.

### 28.3 Self-Attention Visualization

The self-attention response visualizes relationships between positions within the feature representation. Unlike channel attention, which reweights feature channels, self-attention allows spatial positions to exchange information.

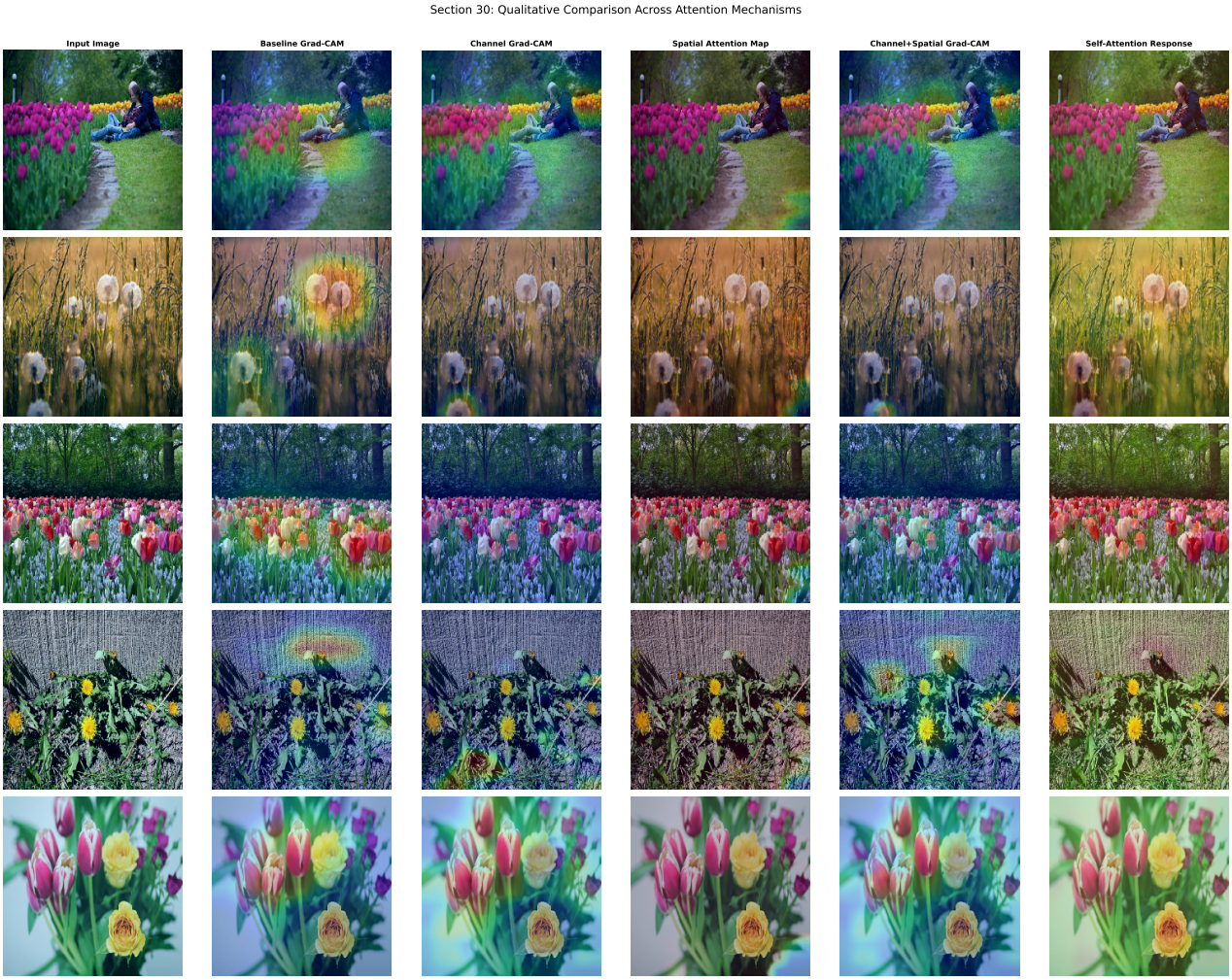
## 29. Attention Position Visualization

The same test image is used to compare spatial attention at different ResNet50 stages.

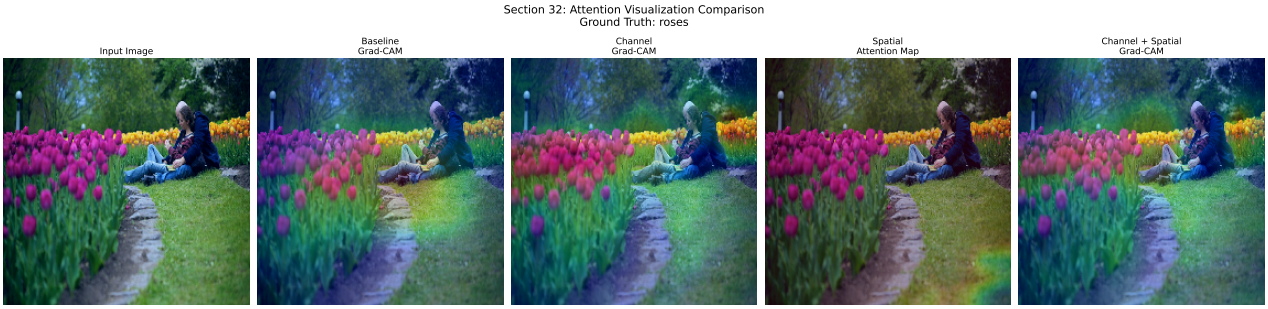
The figure compares $P_2$, $P_3$, and $P_4$. These stages operate on different spatial resolutions and feature representations. Any claim that a stage attends more strongly to edges, flower parts, or complete flower regions should be made only where that behaviour is visible in the figure.

## 30. Qualitative Comparison Across Attention Mechanisms

The notebook-generated qualitative comparison is included below.




The combined visualization presents the input alongside selected model explanations.




The figures should be compared across correctly and incorrectly classified examples. Useful points to inspect include foreground localization, background response, activation coverage, and whether the highlighted region changes when the model prediction is incorrect. The examples are qualitative illustrations and should be interpreted alongside the quantitative test metrics.

## 30.1 Qualitative Criteria

The comparison is based on the following criteria:

- whether activation covers the flower or unrelated background;
- whether petals and the flower centre are highlighted;
- whether activation is compact or spread across a wider region;
- whether incorrect predictions are associated with different highlighted regions; and
- whether the visual behaviour is consistent across multiple examples.

The figures should be used as evidence for the observations; a model should not be described as visually better without pointing to a specific difference in the displayed results.

# 31. Required Experimental Analysis

## 33.1 Quantitative Analysis

The quantitative analysis uses accuracy, precision, recall, and macro F1-score. The recorded attention-type comparison shows that channel attention achieved accuracy 0.9401 and macro F1-score 0.9389.

## 33.2 Complexity Analysis

The complexity analysis considers total parameter count and inference time. The combined-attention model had $25,042,217$ parameters and a recorded inference time of 8.3501 ms/image in the primary attention-type comparison. The baseline had $23,597,957$ parameters and 7.6947 ms/image in that comparison.

## 33.3 Qualitative Analysis

The qualitative analysis uses Grad-CAM, internal attention maps, feature maps, and attention-position visualizations. These provide visual evidence about model responses, but do not replace the quantitative comparison.

# 32. Experimental Protocol

The experiments were designed to reuse the same dataset subsets, preprocessing, backbone, classifier, and evaluation metrics. The attention type or insertion position was the intended variable in each ablation. This controlled setup makes comparisons more meaningful because changes in the reported results are less likely to be caused by different test samples.

The results should still be interpreted as the outcomes of the recorded training runs; small differences may also be affected by initialization, optimization, and runtime conditions.

# 33. Minimum Mandatory Ablation Set

The recorded work includes the baseline ResNet50, channel attention at $P_3$, spatial attention at $P_3$, channel–spatial attention at multiple positions, self-attention at $P_3$, Grad-CAM comparisons, attention maps at different stages, and graphical comparisons of classification performance.

The primary attention-type and position experiments are reported in Sections 14 and 15. The corresponding visual explanations are presented in Sections 27–30.

# 34. Additional Student Exploration

The additional attention-mechanism comparison includes SE, ECA, and CBAM alongside the baseline and the attention mechanisms implemented earlier. Their quantitative results are summarized in Section 19. The qualitative comparisons are included in Sections 27–30.

In the recorded comparison, SE and CBAM did not exceed the baseline accuracy, while channel attention produced the highest accuracy among the primary attention-type models. These findings apply to the reported experimental run.

# 35. Questions Students Must Answer

1. **What is the purpose of residual connections in ResNet50?** Residual connections provide shortcut paths that help information and gradients pass through the network, making deep residual networks easier to optimize.
2. **What does conv2_x represent compared with conv5_x?** The earlier conv2_x stage produces higher-resolution, lower-level features, while conv5_x produces lower-resolution features with more high-level semantic information.
3. **Why does spatial resolution decrease with network depth?** Downsampling reduces the spatial dimensions as the network builds more abstract representations and controls computational cost.
4. **Why does the number of feature channels increase?** More channels allow deeper stages to represent a wider range of learned features and combinations of visual patterns.
5. **What are low-level, intermediate, and high-level features?** Low-level features include edges and colours; intermediate features include local patterns and flower parts; high-level features represent larger object structures and class-discriminative information.

6. **What is the purpose of attention?** Attention reweights features so that some channels or spatial locations have a greater influence on the resulting representation.
7. **What does channel attention learn?** It learns relative importance weights for feature channels.
8. **What does spatial attention learn?** It learns a spatial weight map indicating which locations in a feature map should receive greater emphasis.
9. **How are channel and spatial attention different?** Channel attention weights feature channels, while spatial attention weights spatial locations.
10. **Why can they be used sequentially?** The first module refines the representation along one dimension, and the second refines the resulting features along the other dimension.
11. **Why might channel attention differ at conv3_x and conv5_x?** The two stages have different feature resolutions, channel counts, and semantic content, so their channel weights operate on different representations.
12. **Why can spatial attention be useful at an intermediate stage?** Intermediate features retain spatial detail while also representing meaningful object parts, which can help the module assign location-based weights.
13. **Why is self-attention expensive at high resolution?** It compares spatial positions with one another, creating an attention matrix whose size grows quadratically with the number of spatial positions.
14. **What happens to the self-attention matrix as resolution increases?** If the number of positions is $N = HW$, the attention matrix has size $N \times N$; increasing $H$ or $W$ can therefore increase memory and computation substantially.
15. **How did performance vary with insertion position?** In the recorded position study, $P_4$ had the highest accuracy at 0.9220, followed by $P_3$ at 0.9201, $P_2$ at 0.9111, and $P_1$ at 0.8802.
16. **How can attention at $P_2$ differ visually from $P_4$?** $P_2$ has a higher spatial resolution and may retain finer details, whereas $P_4$ has a lower resolution and represents deeper semantic features. The actual differences should be confirmed from the maps in Section 29.
17. **Does attention at multiple stages always improve performance?** No. In the recorded multi-position experiment, $P_3 + P_4$ reached 0.9383 accuracy, but this does not establish that multiple stages always help or that the extra cost is always justified.
18. **Does attention order matter?** In the recorded comparison, spatial–channel attention reached accuracy 0.9274, compared with 0.9201 for channel–spatial attention.
19. **How does kernel size influence the spatial map?** The kernel changes the receptive field used to compute the spatial map. In the recorded results, $5 \times 5$ and $7 \times 7$ had the same accuracy and macro F1, both slightly above $3 \times 3$.
20. **How does SE differ from ECA?** SE uses a bottleneck transformation to produce channel weights, while ECA uses a lightweight local channel interaction without the same dimensionality-reduction bottleneck.
21. **How does CBAM combine attention?** CBAM applies channel attention and spatial attention sequentially.
22. **How does self-attention differ from channel attention?** Channel attention assigns weights to channels; self-attention models relationships between spatial positions.
23. **Does the model with the most parameters necessarily achieve the highest accuracy?** No. The combined-attention model had more parameters than channel attention but recorded lower accuracy.
24. **What is the difference between an internal attention map and Grad-CAM?** An internal attention map is produced by the attention module during the forward pass. Grad-CAM uses gradients after a prediction to highlight regions relevant to a target class.
25. **Why is Grad-CAM post-hoc?** It is computed after the model has made a prediction, using gradients of the target class score with respect to convolutional features.
26. **Does attention suppress irrelevant background?** This should be determined from the attention overlays and Grad-CAM figures; the numerical accuracy values alone cannot establish it.
27. **Do correct and incorrect predictions have different maps?** The qualitative figures allow this comparison, but the conclusion should be based on the displayed examples rather than assumed in advance.
28. **Which flower classes benefit most from attention?** This should be determined by comparing the class-wise confusion matrices and metrics. Overall accuracy alone does not identify the class that benefits most.
29. **What does a confusion matrix show beyond accuracy?** It shows the specific true and predicted class pairs, including which classes are confused with one another.
30. **Why use macro F1 in addition to accuracy?** Macro F1 averages class-wise F1-scores and therefore gives each class equal weight, helping reveal uneven class-wise performance.
31. **What is the accuracy–complexity trade-off?** It compares classification performance with the parameters and computation needed to obtain that performance.
32. **What is the F1–inference-time trade-off?** It considers whether any increase in macro F1 is meaningful relative to the additional time required to process each image.
33. **Why use the same dataset split?** Using the same samples prevents the comparison from being confounded by differences in training, validation, or test data.
34. **Why maintain the same training protocol?** Keeping the protocol fixed makes the intended change in attention type or position the main factor being compared.
35. **What conclusions can be drawn from Grad-CAM?** Grad-CAM shows which spatial regions contribute to

a class score in the selected examples. It should be interpreted together with the input, prediction, attention maps, and quantitative results.

## 36. Observations to be Recorded

The principal recorded values include test accuracy, macro precision, macro recall, macro F1-score, total parameter count, and average inference time. The main results are summarized in Section 19.1. Feature dimensions before and after the $P_3$ attention block were recorded as $14 \times 14 \times 1024$, confirming that the block did not change the feature tensor shape.

The report also includes the confusion matrices, training and validation curves, channel-attention weights, spatial-attention maps, self-attention response, Grad-CAM figures, and attention maps for different insertion positions.

## 37. Suggested Laboratory Report Structure

The report follows the order and topics specified in the laboratory manual. Sections 1–13 introduce the objective, dataset, ResNet50 feature hierarchy, and attention mechanisms. Sections 14–19 report the attention ablation experiments and their measured results. Sections 20–25 describe the transfer-learning setup, evaluation metrics, result tables, graphs, and confusion matrices. Sections 26–32 present the qualitative visualizations, while Sections 31–38 summarize the analysis, protocol, and recorded observations.

## 38. Discussion Guidelines

The recorded experiments show that attention performance depends on both the mechanism and the feature stage at which it is inserted. In the primary attention-type comparison, channel attention achieved the highest accuracy (0.9401) and macro F1-score (0.9389), compared with baseline accuracy 0.9347. The channel–spatial model at $P_3$ did not exceed the baseline in that run.

For the position experiment, $P_4$ had the highest accuracy among the four tested positions, while $P_1$ had the lowest. The multi-position experiment reported accuracy 0.9383 for $P_3 + P_4$, higher than the single-position results in that comparison. The attention-order experiment favoured spatial–channel order, and the $5 \times 5$ and $7 \times 7$ kernel results were equal on the reported accuracy and macro F1 metrics.

These findings should be considered with model complexity and inference time. The qualitative figures provide additional information about model responses, but statements about flower localization or background suppression must be supported by the actual visual evidence. Taken together, the numerical and visual results provide a more complete assessment than accuracy alone.

## 39. Expected Outcome

The experiment demonstrates that the effectiveness of attention depends on the attention type, insertion position, feature resolution, feature semantics, and model complexity. In the recorded attention-type comparison, channel attention gave the strongest accuracy and macro F1-score, while other attention configurations produced different results.

The report combines quantitative metrics, computational measurements, and qualitative visualizations. The results do not establish a universally best attention mechanism; they show which configurations performed best under the recorded experimental setup.

$$\boxed{\text{Complete Attention Analysis} = \text{Quantitative} + \text{Computational} + \text{Qualitative}}$$

## 40. References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4. Q. Wang, B. Wu, P. Zhu, P. Li, W. Zuo, and Q. Hu, "ECA-Net: Efficient Channel Attention for Deep Convolutional Neural Networks," *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 11534–11542, 2020.
5. Q. Hou, D. Zhou, and J. Feng, "Coordinate Attention for Efficient Mobile Network Design," *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 13713–13722, 2021.
6. A. Vaswani et al., "Attention Is All You Need," *Advances in Neural Information Processing Systems*, vol. 30, 2017.
7. R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization," *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pp. 618–626, 2017.
8. TensorFlow Documentation: https://www.tensorflow.org/
9. Keras Documentation: https://keras.io/
10. TensorFlow Datasets: https://www.tensorflow.org/datasets

## Source Code

The complete implementation is available in the GitHub repository.

**GitHub Repository:**

https://github.com/KPR-24110081/Deep-Learning-Lab/tree/main/Ex-8,9